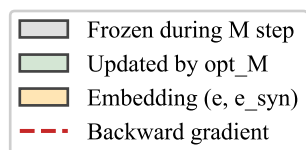


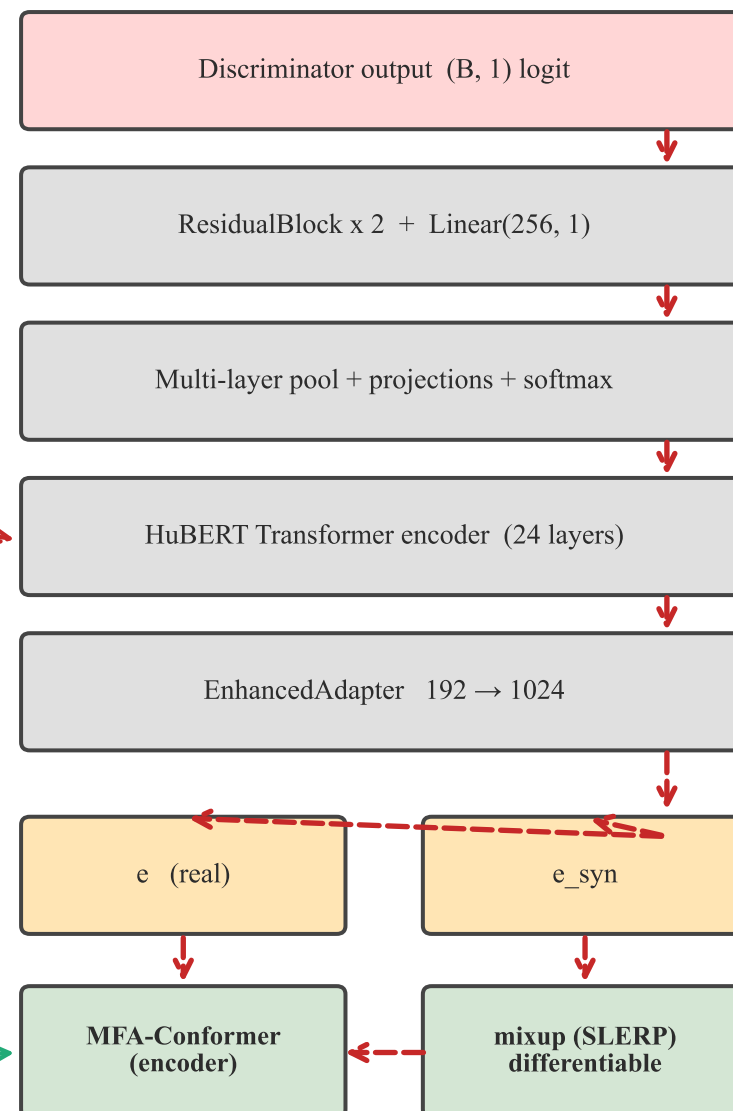
Gradient flow during the M step (encoder update)



All of D is frozen — but the gradient still flows through it

These layers are FROZEN
(requires_grad = False)

Parameters are NOT updated by
opt_M, but the gradient STILL
TRAVERSES the computation
graph all the way to the inputs.



Gradient arrives here:
only MFA-Conformer params
and AM-Softmax W are updated
by opt_M.step()

Key insight: `requires_grad = False` blocks parameter UPDATES, NOT gradient propagation.
This is precisely how a frozen HuBERT still trains the encoder through D.